Fixed-Byte Source-Private Candidate Packets for Bounded Model Collaboration

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Abstract

Multi-model language systems usually communicate by emitting text or by sharing internal state such as KV caches. Text is portable but slow and lossy; cache sharing is information-rich but exposes thousands of bytes of model state per token and requires architecture-specific machinery. We study a narrower question: can a frozen source model send a fixed-byte, source-private candidate packet that improves a frozen receiver on public multiple-choice reasoning tasks, without sending source text, hidden states, KV cache, source logits, or gold labels? We present LatentWire, a public-coordinate packet protocol in which source and receiver agree on an anchor/spectral basis over candidate answers, the source emits a tiny signed packet, and the receiver decodes the packet against its own public side information. On ARC-Challenge test, an 8-byte packet reaches 0.344 accuracy across 10 seeds, above target-only 0.265 and same-byte structured text 0.300, with destructive coordinate controls failing in all 10 seeds; the source-choice accuracy is 0.346, so the positive should be read as compact source-candidate transfer rather than evidence synthesis beyond the source. On OpenBookQA test, a 3-byte packet reaches 0.378 across 5 seeds, above target-only 0.276 and same-byte text 0.350, matching the source-choice accuracy. We also report strict negative evidence: a Phi-3 cross-family source fails the same gate, and learned cache/score connectors built from available cached artifacts do not repair it. The contribution is therefore not a universal latent language claim and does not beat explicit source-index communication. It is a reproducible fixed-byte source-private transfer result, a falsification protocol for cross-family claims, and a systems accounting boundary showing that task packets are 69.8x smaller than even a one-token 1-bit-per-KV-element accounting floor.

1 Introduction

Future language-model systems will often use more than one model: a cheap model may screen candidates, a specialist may inspect hard cases, and a stronger receiver may combine evidence. The simplest interface is text. Recent cache-level systems instead communicate through model internals, such as fused or selectively shared KV caches (Fu et al., 2026; Shi et al., 2026). These approaches are powerful, but they couple communication to large hidden-state objects and serving substrates.

This paper asks whether a much smaller interface can carry useful source-side candidate evidence. In our setting, both models see the same public question and answer choices. The source performs private computation over those choices. The only communicated object is a short byte packet; the receiver never sees source text, source logits, source hidden vectors, KV cache, or the answer key. This is a side-information problem in the spirit of distributed source coding (Slepian & Wolf, 1973; Wyner & Ziv, 1976): the receiver already has the public prompt and its own computation, so the packet should carry only the source evidence that is useful at the decoder.

Our strongest current result is deliberately scoped. On ARC-Challenge and OpenBookQA (Clark et al., 2018; Mihaylov et al., 2018), same-family Qwen2.5 packet transfer (Yang et al., 2025) improves over target-only and same-byte text controls. However, a

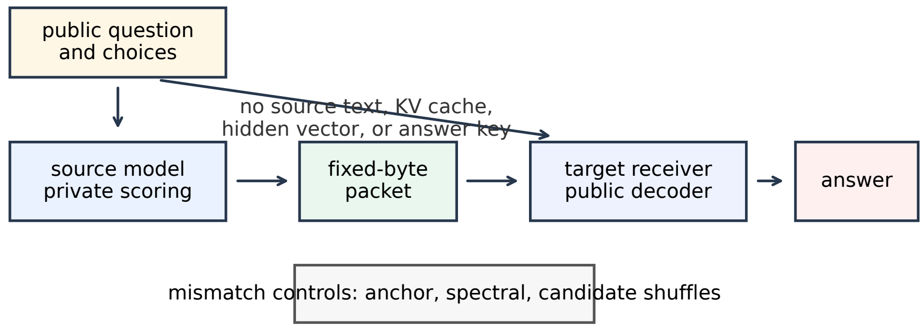

Figure 1: LatentWire protocol. Both sides receive the public task. The source privately scores candidates and emits only a fixed-byte packet. The receiver decodes the packet with a public basis and its own side information. Destructive controls scramble the shared coordinate system or candidate relation.

strict Phi-3 source-family replacement (Abdin et al., 2024) fails, and additional connector repairs fail on the available cached artifacts. This boundary matters: the paper is not claiming solved cross-family latent communication.

The contributions are:

- A fixed-byte, source-private packet protocol for multiple-choice candidate transfer with no source text, hidden vector, KV cache, source-logit, or gold-label exposure.
- A public-coordinate anchor/spectral implementation that is positive on ARC-Challenge and OpenBookQA under seed repeats, paired uncertainty, same-byte text controls, and destructive coordinate mismatch controls.
- A source-choice and falsification audit showing that the current positive is mostly source-choice preserving and same-family/source-cache specific: Phi-3 cross-family transfer and cached learned repair connectors fail.
- A systems accounting artifact comparing packet bytes and exposure classes against KV/cache communication and KV quantization floors, while explicitly separating byte accounting from native serving-speed claims.

2 Source-private packet transfer

Let a public multiple-choice instance be $x = (q, c_1, \ldots, c_m)$ with answer choices c_i. A source model S and receiver R both observe x, but only S computes the source-side private evidence $e_S(x)$. A packet encoder emits

$$z = E(e_S(x), x), \qquad |z| \leq B \text{ bytes.}$$

The receiver predicts

$$\hat{y} = D_R(x, z),$$

where D_R may use public task features and the receiver's own state, but may not access source text, source hidden states, source KV cache, source logits, or answer labels.

We evaluate a packet only if it beats three surfaces: target-only $D_R(x, \emptyset)$, a same-byte structured text control, and destructive controls that preserve superficial packet size but destroy the intended source-receiver relation. We report seed repeats and paired bootstrap lower bounds, and we separately report source-choice accuracy where available. This matters because a packet can be useful while mostly preserving the source's selected candidate; such a result is source-private transfer, not proof of a shared latent language or of reasoning beyond the source endpoint.

Benchmark	split	bytes	seeds	source	packet	target	text	CI low
ARC-Challenge	test, 1172	8/11	10/10	0.346	0.344	0.265	0.300	+0.038
OpenBookQA	test, 500	3/6	5/5	0.378	0.378	0.276	0.350	+0.038

Table 1: Main positive rows. "Bytes" reports payload/framed bytes. "Source" is the answer-key-forbidden source-choice accuracy before packet decoding. "Text" is the same-byte structured text control. "CI low" is the minimum paired-bootstrap 95% lower bound for packet minus target-only.

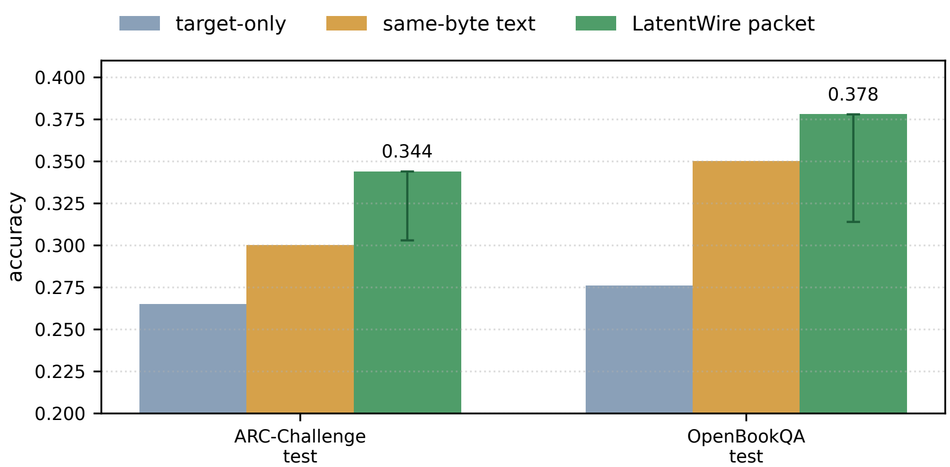

Figure 2: Packet accuracy against target-only and same-byte structured text controls. Error bars show the lower paired-bootstrap lift above target-only; upper uncertainty is omitted for readability because the gate is one-sided.

3 Public-coordinate packets

The current ARC-Challenge implementation uses a public train-anchor chart. For each candidate, we compute a public relative-coordinate feature against a frozen set of training anchors, then project that feature into a low-frequency orthonormal DCT-II spectral basis. The packet is a sparse signed syndrome in this shared coordinate system. The receiver scores candidate compatibility with the packet in the same public basis, combining the packet evidence with receiver-side candidate information.

The design is related to relative representations (Moschella et al., 2023): anchors provide a coordinate system less tied to raw hidden dimensions. Our use is narrower. We do not claim zero-shot latent stitching or global representation alignment; we use a public basis to define a tiny task packet. Random shared anchors work nearly as well as semantic anchors in the current hashed ARC-Challenge implementation, so the correct interpretation is shared coordinate agreement, not semantic anchor names.

OpenBookQA uses the same packet-evaluation contract at a smaller 3-byte budget. Across both tasks, the source-choice cache is answer-key-forbidden: the source sees public questions and choices, not labels. The packet is evaluated on frozen test slices after validation selection.

4 Main results

Table 1 and Figure 2 summarize the current positive evidence. The ARC-Challenge row uses 10 packet/control seeds and 2000 paired bootstrap samples per seed on 1172 test examples. The OpenBookQA row uses 5 seeds on the 500-example test slice.

Source	surface	n	packet	target	text	ref. pkt	blocker
Phi-3	full test	1172	0.244	0.265	0.232	–	source quality
Phi-3	Qwen-disagree	833	0.200	0.273	0.203	0.340	source quality
TinyLlama	Qwen-disagree	473	0.269	0.268	0.258	0.317	family mismatch
Qwen2.5-1.5B	full test	1172	0.442	0.265	0.401	–	val incomplete

Table 2: Cross-family and source-family diagnostics. "Ref. pkt" is the Qwen-substituted packet on disagreement rows. Qwen2.5-1.5B is encouraging but same-family and validation-incomplete, so it is not a headline cross-family claim.

On ARC-Challenge, the matched Fourier/anchor-syndrome packet reaches 0.344 mean test accuracy with minimum seed accuracy 0.342. The answer-key-forbidden source choice itself reaches 0.346, and packet predictions match the source-selected candidate at 0.995 mean rate across seeds. Anchor-ID shuffle, anchor-value shuffle, and spectral-bin permutation controls all fail in 0/10 seeds and collapse near target-only. The random shared-anchor diagnostic passes, which weakens any claim that named anchors are semantically special, but strengthens the claim that a shared public coordinate chart is the relevant mechanism.

On OpenBookQA, the 3-byte packet reaches 0.378–0.380 across seeds, matching the source-choice accuracy of 0.378. The largest candidate-derangement control accuracy is 0.212, and the minimum lift over same-byte structured text is +0.028. This second task matters because it reduces the risk that the result is only an ARC-Challenge artifact, though it is still a same-family packet result rather than a cross-family result.

5 Falsification and failure decomposition

The most important negative result is the Phi-3 source-family replacement. We keep the same ARC-Challenge packet budget, public basis, seed count, and bootstrap budget, but replace the Qwen source-choice cache with a Phi-3 source. The packet no longer passes. On the full ARC-Challenge test slice, the Phi-3 packet reaches 0.244 versus target-only 0.265. On the stricter Qwen-disagreement slice, where the alternate source and Qwen source disagree, the Phi-3 packet reaches 0.200 versus 0.340 for the Qwen-substituted packet.

The failure decomposition separates three causes: source endpoint quality, packet corruption, and common-feature mismatch. For Phi-3, packet predictions follow the Phi-3 source choice at 0.997, but the source itself is below the target. For TinyLlama, packet fidelity is also high, but the source-family mismatch causes the disagreement-slice packet to lose to Qwen-substituted packets. For Qwen2.5-1.5B, the source is stronger and the packet is useful on frozen test, but the validation disagreement gate is incomplete. Across these rows, the packet mostly transports the source-selected candidate through the public coordinate basis. The next research gate is therefore not another scalar packet shuffle; it is either an explicit comparison to source-index/score packets or a richer common-feature connector that can beat packet-only source choice.

We also tested cached learned repairs on the available local artifacts. A candidate-syndrome connector that sees packet and score-shape features reaches 0.288 versus 0.317 for Qwen-substituted packets on frozen TinyLlama/Qwen disagreement rows. A one-hidden-layer MLP over cached TinyLlama hidden/query summaries reaches 0.232, below Qwen-substituted packets, cached Tiny packets, target-only, and same-byte text. These negatives are useful: simple cached score geometry and mean hidden/query summaries do not explain the remaining oracle headroom.

6 Systems boundary

The systems contribution is byte/exposure accounting, not native serving throughput. Figure 3 compares framed packet bytes with one-token KV/cache-state accounting floors from neighboring systems and quantization methods. LatentWire sends 6–11 framed bytes

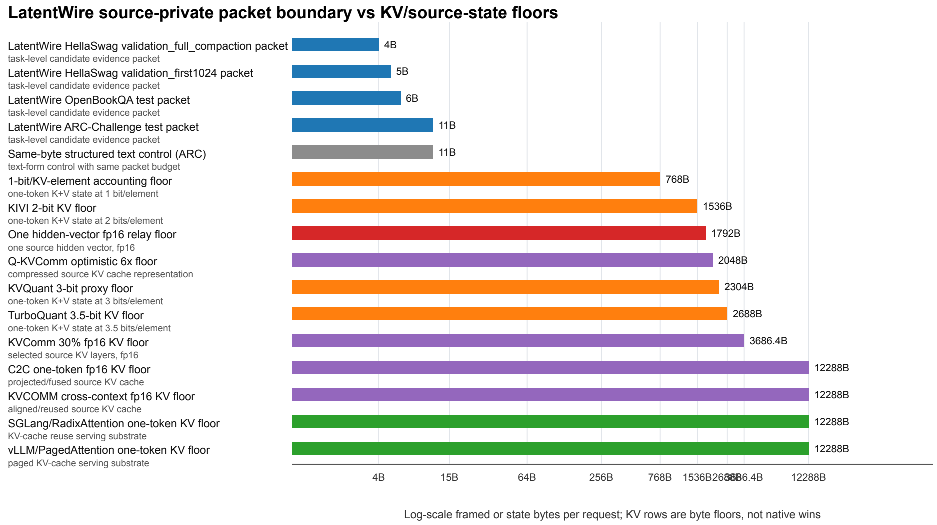

Figure 3: Boundary-byte accounting. Packet rows are measured task-packet objects. KV/cache rows are lower-bound floors or neighboring serving substrates, not native speed comparisons.

Method	raw	framed	text	KV/hidden	claim status
LatentWire OpenBookQA packet	3B	6B	no	no	local positive
LatentWire ARC-Challenge packet	8B	11B	no	no	local positive
Same-byte text control	8B	11B	yes	no	control only
1-bit/KV-element floor	768B	768B	no	KV	floor only
KIVI 2-bit KV floor	1536B	1536B	no	KV	floor only
C2C one-token KV floor	12288B	12288B	no	KV	native pending

Table 3: Exposure classes. The paper-ready systems claim is that the communicated object is much smaller and less exposing than KV/cache-state objects under this accounting. It is not a GPU throughput, TTFT, TPOT, or HBM claim.

on the two headline rows. A one-token 1-bit-per-KV-element accounting floor is 768 bytes in our model-configuration accounting; the largest headline packet is therefore 69.8x smaller than that floor. C2C-style cache transfer, KVComm-style selective sharing, and serving substrates such as vLLM/PagedAttention and SGLang operate on KV/cache objects rather than task-level packets (Fu et al., 2026; Shi et al., 2026; Kwon et al., 2023; Zheng et al., 2024). KV quantization work reduces cache bytes but still communicates or stores model state (Zandieh et al., 2024; Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2026).

7 Related work

Latent and relative representation communication. Relative representations use similarities to anchors to enable latent-space communication under invariances (Moschella et al., 2023). Sparse autoencoders and feature-space universality work motivate common-feature interfaces (Huben et al., 2024; Lan et al., 2024). SAEBench provides evaluation machinery for sparse autoencoders (Karvonen et al., 2025), but these works do not by themselves provide a packet protocol or task-level evidence transfer result.

Prompt compression and learned prefixes. Prefix tuning and gist tokens learn continuous or discrete prompt-like interfaces within language models (Li & Liang, 2021; Mu et al., 2023). LatentWire differs because the communicated object is a byte packet decoded by a public receiver, not a learned soft prompt injected into one target. Query-bottleneck connectors such as BLIP-2's Q-Former show that learned bottlenecks can bridge frozen model families across modalities (Li et al., 2023); our failed cached hidden/query summaries suggest that a true tokenwise query-bottleneck is the right future branch, but it is not closed here.

Cache communication and systems. Cache-to-cache and selective KV-sharing systems communicate richer internal state and are the closest competitors for model-to-model communication (Fu et al., 2026; Shi et al., 2026). vLLM and SGLang improve serving through KV-cache management and reuse (Kwon et al., 2023; Zheng et al., 2024). KV quantizers such as QJL, KIVI, KVQuant, and TurboQuant reduce state cost but remain state-transfer or state-storage techniques (Zandieh et al., 2024; Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2026). Our current systems result is a boundary artifact against these state objects, not a native throughput win.

8 Limitations and next gate

The main limitation is scope. The positive results are same-family/source-cache results on public multiple-choice tasks, and the packet largely preserves the source's selected candidate. We therefore do not claim to beat explicit source-index communication, source-score quantization, or calibrated source-choice baselines; those are required before a stronger compression claim. Phi-3 cross-family replacement fails, TinyLlama common-feature repair fails, and a true cross-family learned query-bottleneck remains future work. We also do not claim superiority over C2C, KVComm, QJL, KIVI, TurboQuant, vLLM, or SGLang in native serving metrics. Those rows require NVIDIA hardware and matched implementations.

The next exact gate for an ICLR-strength claim is a frozen-endpoint tokenwise connector with an explicit byte or query budget, tested against C2C/KV-style and structured-text baselines, with source-destroying controls and strict same-family versus cross-family separation. The current strongest claim is narrower: fixed-byte source-private packets can transfer useful same-family evidence on ARC-Challenge and OpenBookQA, and the accompanying falsification ladder clarifies where the method stops working.

9 Conclusion

LatentWire shows that a tiny source-private candidate packet can improve frozen receiver decisions on two public QA benchmarks under strict same-family controls. The same evidence also draws a hard boundary: the current method is mostly source-choice preserving, is not a universal latent language, and does not yet solve cross-family communication. This combination of positive packet transfer, negative falsification, and systems byte/exposure accounting is the paper's core.

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A Artifact manifest

- ARC-Challenge headline: `results/source_private_arc_challenge_fourier_anchor_syndrome_gate_20260502_budget8_10seed_b2000`.
- Phi-3 falsification: `results/source_private_arc_challenge_source_family_cache_falsification_20260502_phi3_cpu_budget8_10seed_b2000`.
- Cross-family decomposition: `results/source_private_arc_cross_family_failure_decomposition_20260502`.
- Systems boundary: `results/source_private_systems_boundary_figure_table_20260502`.
- OpenBookQA headline: `results/source_private_openbookqa_seed_stability_20260501_qwen05_hashed_test_3b`.

B Reproducibility summary

The artifact bundle records exact shell commands, input hashes, output hashes, and rerun results. The paper rows are produced from cached, answer-key-forbidden source-choice artifacts with the following settings:

Row	script	seeds	key settings
ARC-Challenge	Fourier/anchor syndrome gate	10	8B, 2000 bootstraps
OpenBookQA	seed-stability gate	5	3B, hashed features, 500 bootstraps
Phi-3	source-family cache falsification	10	8B, cached Phi-3 source
Systems	systems boundary table	–	byte/exposure accounting only

The reruns are metric-reproducible, but not intended to be byte-identical: JSON files contain creation times and several artifacts include measured local latencies.

C Lay explanation of the main experiment

The experiment asks whether one model can send a tiny hint to another model. Both models see the same question and answer choices. The sender chooses a compressed coordinate hint, only a few bytes long. The receiver uses that hint to choose an answer. If both sides agree on the coordinate system, the hint helps. If we scramble the coordinate system, the hint stops helping. That is evidence for a real shared packet interface on the tested same-family setting, while the failed Phi-3 run shows the interface is not yet general across model families.